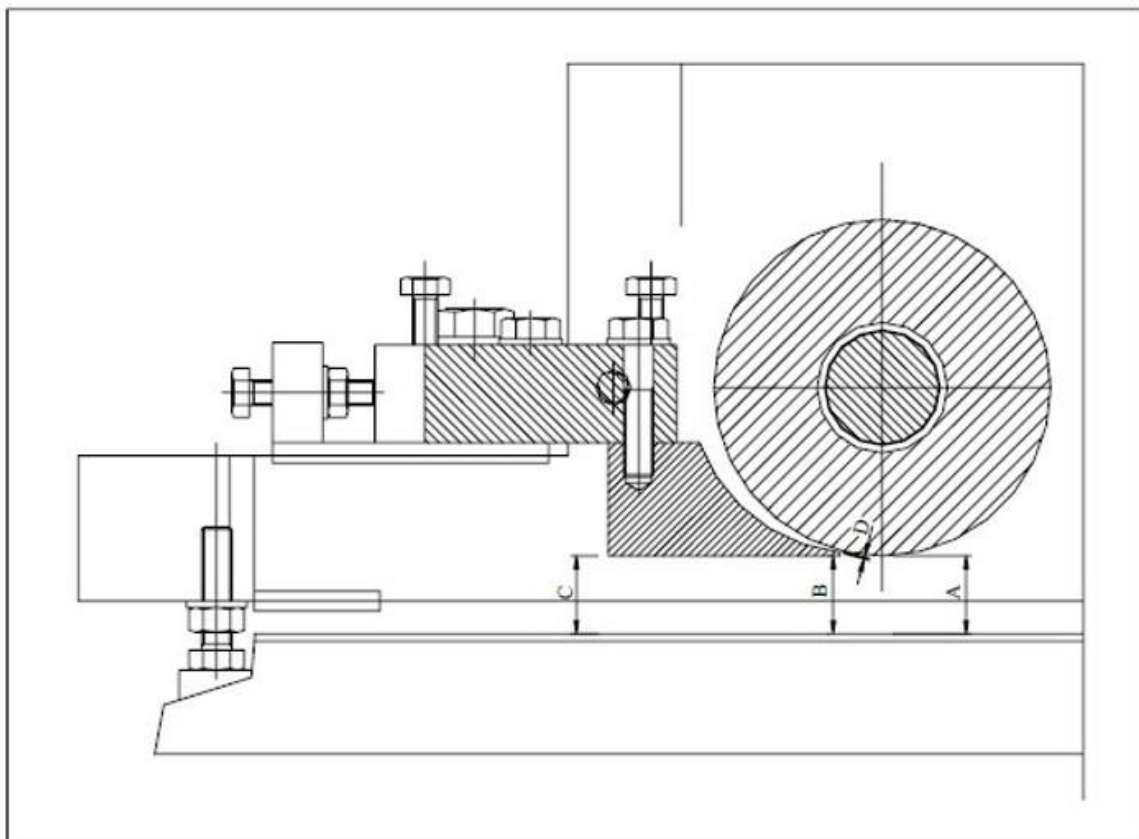


EXTRUSION HEAD / ROOF TILE FORMING

The TILE SHAPPING PROCESS that I am going to present consists of 16 indicators that I consider "necessary to take into account", in order to control the whole "Tile Shaping Process".

In the development of this process, each factory can expand or reduce the proposed indicators as they see fit.

There are 4 main indicators in this Process, which are the measures of adjustment of the tile head, **A**, **B**, **C** and **D**, they are very particular for each model of tile and conditioned to the raw materials that are used for the manufacture of the tile.



The measurements "**A**", "**B**", "**C**" and "**D**" correspond to:

"**A**" = Distance from the bench guides to the flat areas of the roller, according to the thickness of the tile in the flat area minus the height of the mould.

"**B**" = Distance from the bench guides to the Big Slipper entrance.

"**C**" = Distance from the bench guides to the exit of the Big Slipper.

("Track slipper inclination mm" = $B - C$)

"**D**" = Distance from the Big slipper to the Big Roller "**GAP D**".

Initially for measures **B, C and D**, there are reference values, as detailed in the table below, always depending on the type of sand (fine / coarse) used.

SLIPPER REFERENCE DISTANCES			
Ref.	Distance	Type of sand	Initial value
A	Distance of roller + pallet	/	10 mm
B	C + 0.8 ÷ 1 mm	With fine sand	/
B	C + 1 ÷ 1.3 mm	With coarse sand	/
C	A + 0.8 ÷ 1 mm	With fine sand	/
C	A + 1.1 ÷ 1.3 mm	With coarse sand	/
D	2 ÷ 3 mm	/	/

There is also an initial reference roller speed, the calculation values are the speed (tiles/minute) and the smallest roller diameter of the tile profile.

However, once in production with the other settings, it can vary, any excess of the roller speed will result in increased wear of the roller

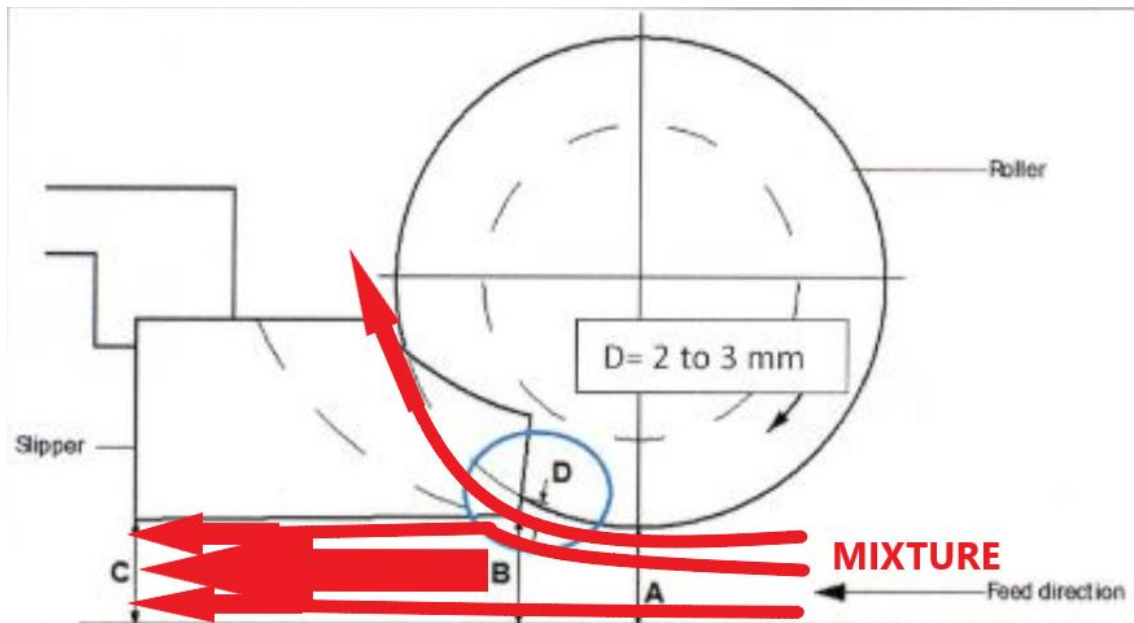
$$\text{Roller Speed (Rpm)} = \frac{(\text{N. Tiles/Min}) * 42 * 1.2}{(\text{D min. Roller in cm}) * 3.14} = \text{R.P.M.}$$

So far, 6 indicators of the Process have been presented.

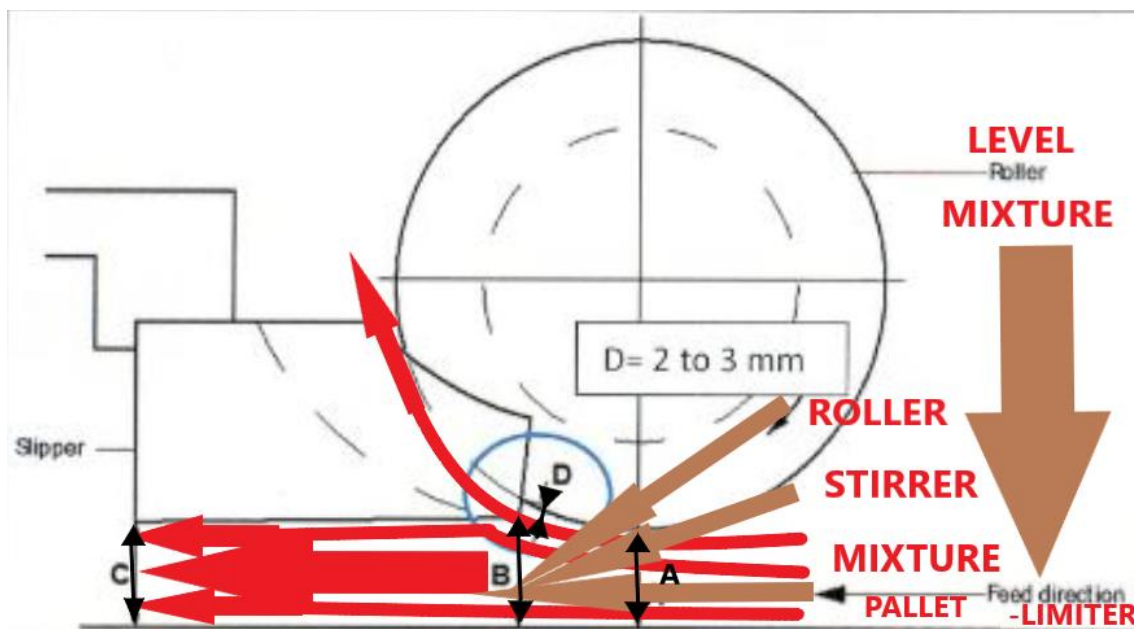
PROCESS SHEET							Code
							Date: 2022
							Review number: 1
							Page: 1
FACTORY:							
SECCION : Automatic extruder E4S 120 tpm							
PROCESS : EXTRUSION HEAD / TILE CONFORMATION							
PROFILE	Nº	PARAMETERS	NOMINAL VALUE	MINIMUM VALUE	MAXIMUM VALUE	UNITS	NOTES
STEEL HEAD OR TUNGSTEN, DOUBLE ROMAN	1	Roller speed.				r.p.m.	ALL THESE VALUES
	2	Stirrer speed.(units / dimensions)				r.p.m.	ARE ESTABLISHED FOR
	3	Roller height. (A)				mm	PRODUCTION SPEED OF
	4	Slipper entry height (B)				mm	TILES PER MINUTE
	5	Slipper exit height. (C)				mm	
	6	Distance from the big slipper to the big roller.GAP (D)				mm	
	7	Limiters length . (steel / polycord))				mm	
	8	New roller diameter.				mm	
	9	Used roller diameter.				mm	
	10	Height distance between big and small slippers				mm	
	11	Maximum roller temperature				°C	
	12	Maximum slipper temperature				°C	
	13	Roller ammeter				Ampere	
	14	Pushing chain ammeter				Ampere	
	15	Column height.				mm	
	16	Tiles per minute				t.p.m	

To understand a little more about the shaping of the tile on a mould, the philosophy is to overfeed concrete into the space between the mould and the roller / slipper, to maintain a fresh concrete on the pallet at a certain compaction, so that it does not deform during the whole process of cutting, moving and stacking.

All the concrete from the hopper supported on the base of the pallet, is fed by the mould itself towards the roller assisted by the rotation of the roller, in that direction with an initial leakage of concrete between the roller and slipper.

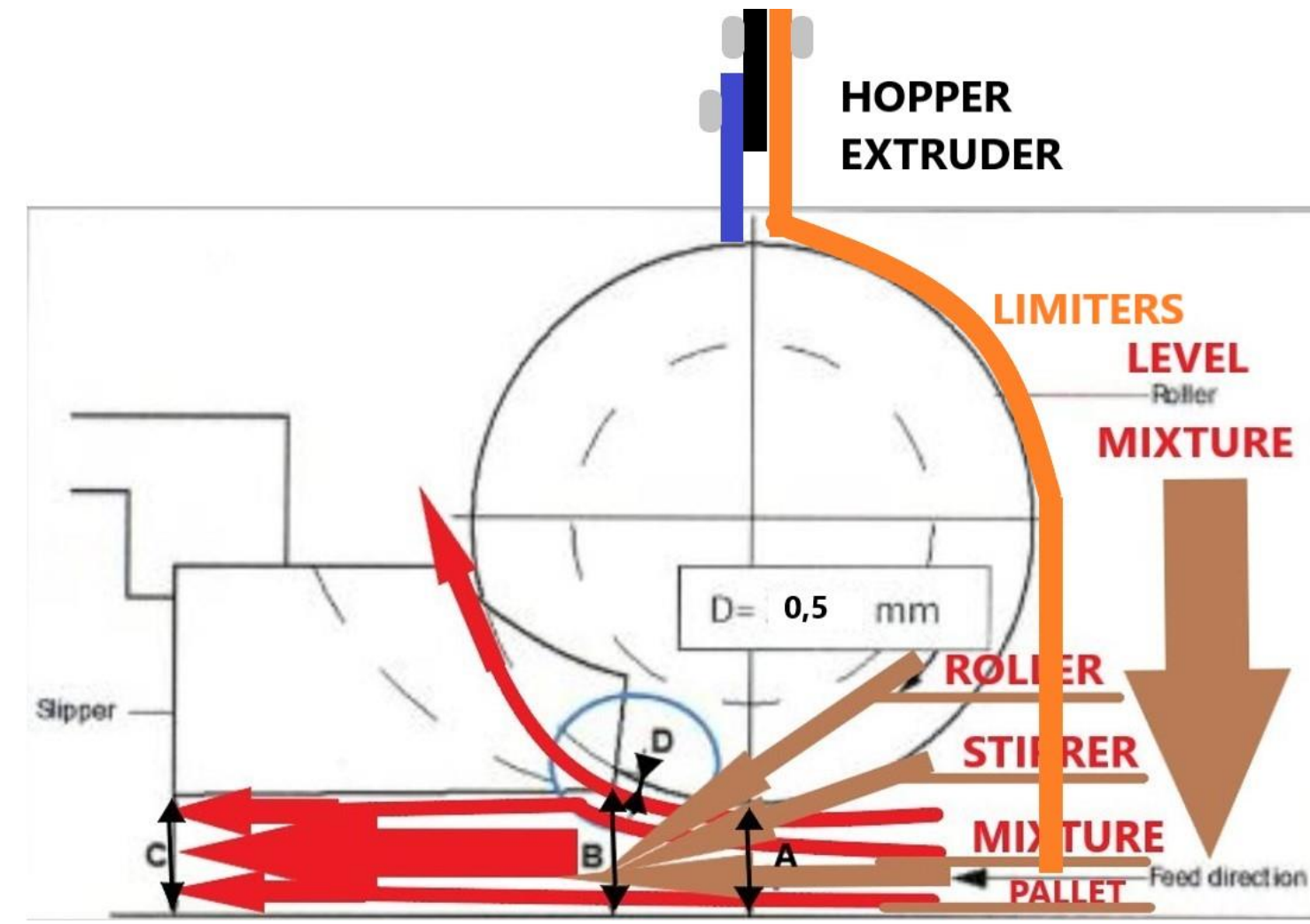


At this point there are already more process indicators, which intervene for this shaping, the height of the mixture column in the hopper (mm), the speed of the roller (rpm), units / dimensions of the stirrers and their speed (rpm), mixture limiters in the flat areas of the tile (YES / NO) (length mm).



		PROCESS SHEET						Code		
								Date: 2022		
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FACTORY:										
SECCION : Automatic extruder E4S 120 tpm										
PROCESS: EXTRUSION HEAD / TILE CONFORMATION										
PROFILE	N°	PARAMETERS	NOMINAL VALUE		MINIMUM VALUE		MAXIMUM VALUE		UNITS	NOTES
STEEL HEAD OR TUNGSTEN, DOUBLE ROMAN	1	Roller speed.							r.p.m.	ALL THESE VALUES ARE ESTABLISHED FOR PRODUCTION SPEED OF TILES PER MINUTE
	2	Stirrer speed.(units / dimensions)							r.p.m.	
	3	Roller height. (A)							mm	
	4	Slipper entry height (B)							mm	
	5	Slipper exit height (C)							mm	
	6	Distance from the big slipper to the big roller.GAP(D)							mm	
	7	Limiters length . (steel / polycord))							mm	
	8	New roller diameter.							mm	
	9	Used roller diameter.							mm	
	10	Height distance between big and small slippers							mm	
	11	Maximum roller temperature							°C	
	12	Maximum slipper temperature							°C	
	13	Roller ammeter							Ampere	
	14	Pushing chain ammeter							Ampere	
	15	Column height.							mm	
	16	Tiles per minute							t.p.m	

LIMITERS

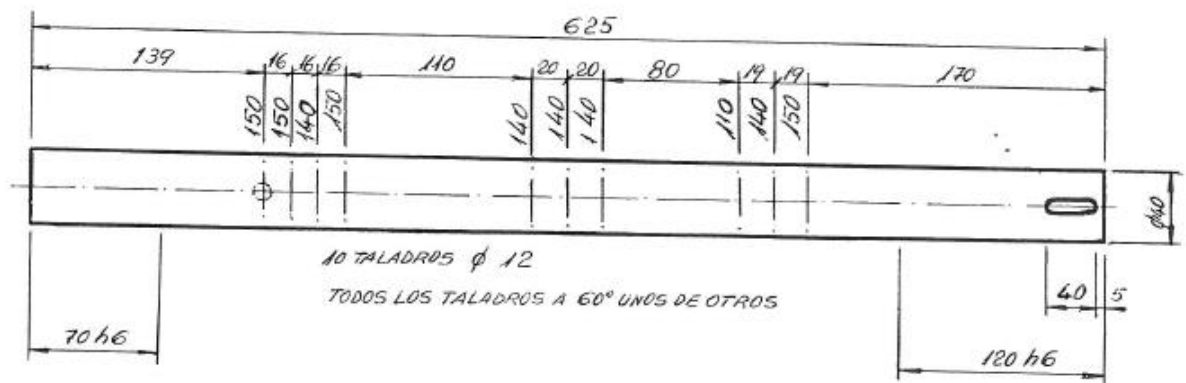


STIRRERS

The stirrer pins are responsible for feeding concrete in all areas of the tile when the gravity of the concrete column of the hopper, the surface of the aluminum pallet and the rotation of the rollers drag the concrete under the slippers, usually its greater the need is in the less thickness areas of the concrete tile.

STIRRER PINS SHAFT PROFILES

“VENETIAN” PROFILE

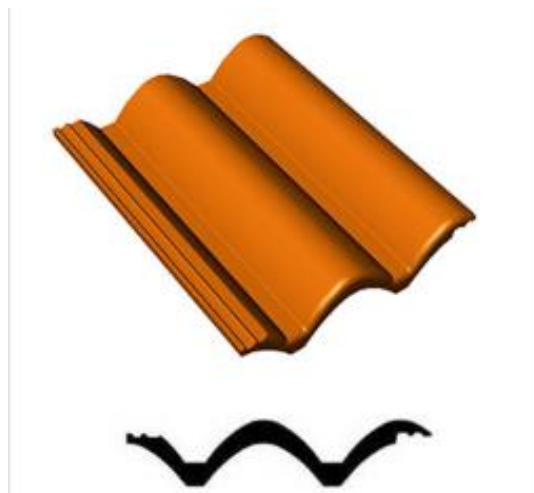


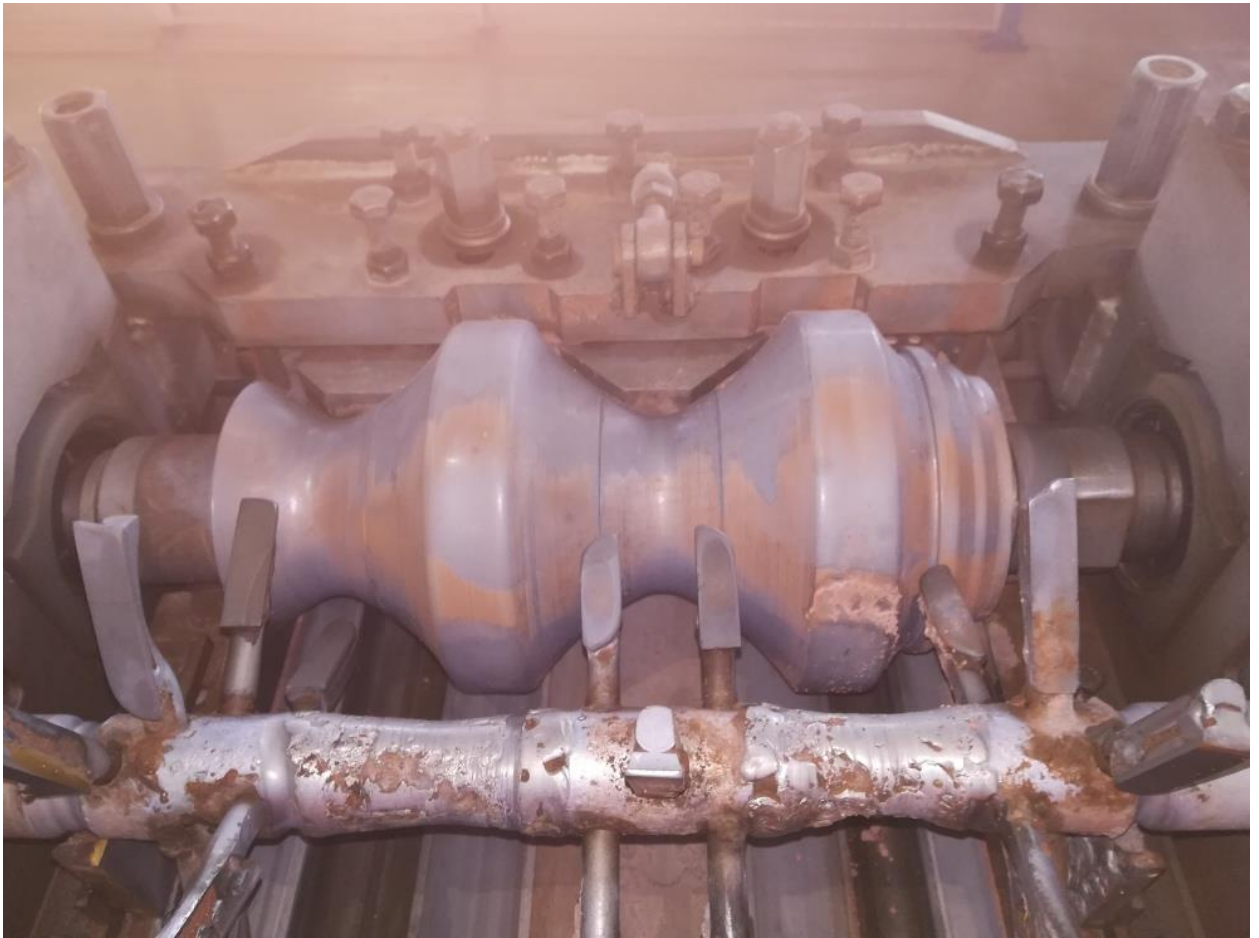
Singularities in the 2 factories for this profile:

It is the stirrer pins shaft with the highest profile and the most demanding need for concrete.

10 units. 12 mm. Star pins diameters. At 60° among them.

Partial sum of the distances in all the holes gives 328 mm, which we only leave 1 mm on each side of the plates, since it corresponds to 330 mm



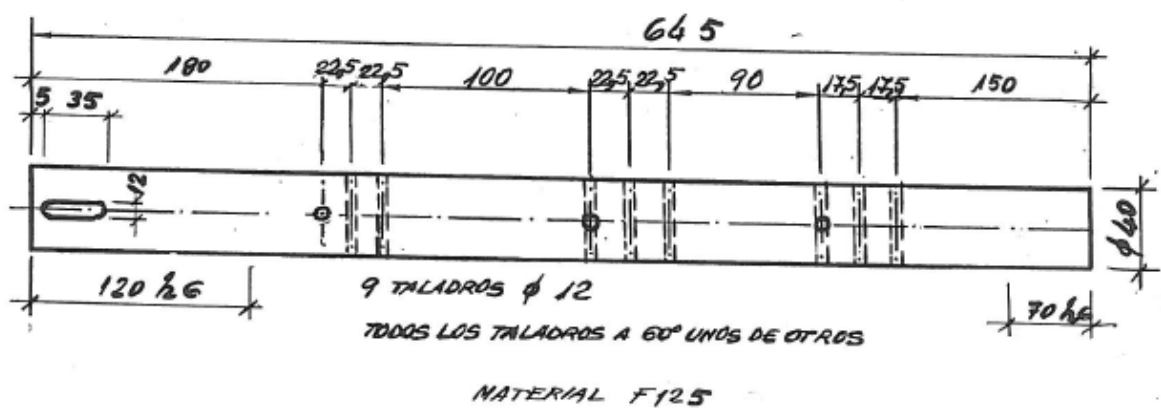


TPM High Speed Extruder Head, "Venetian" tile profile with 10 in-line star pins.

Stirrer pins with diameter 8 mm, widia tablets welded, shaft coated tungsten carbide to prevent wear..

In this extrusion head with a different philosophy, much feed for input by star pins. Vertical limiters

"ARCADIA" PROFILE

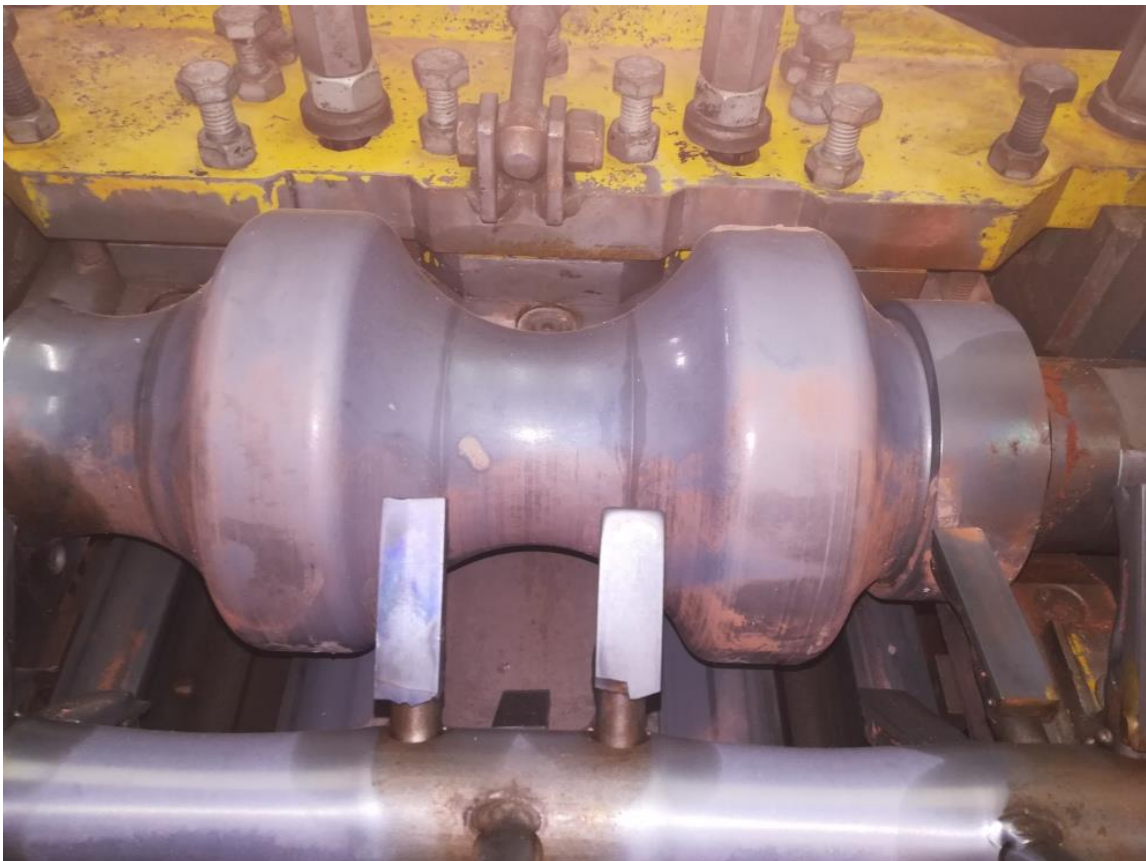
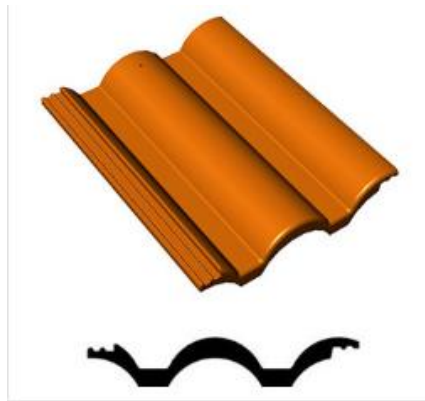


Singularities in the 3 factories for this profile:

It is the stirrer pins shaft with a smaller profile than the “Venetian”, with the wider planes and less need for power.

9 units. 12 mm. Stirrer pins diameters. At 60° among them.

Partial sum of the distances in all the holes gives 327 mm, which we only leave 1.5 mm on each side of the plates, since it corresponds to 330 mm.



TPM High Speed Extrusion head, “Arcadia” tile profile with 9 in-line stirrer pins.
Stirrer pins with diameter 12 mm, width tablets.

In this extrusion head with a different philosophy, much feed for input by stirrer pins.
Vertical limiters

Summary:

In the examples raised on stirrer pins there is a quantity, distribution and angles between them (90° or 60°) for each profile depending on the needs that may have on the concrete feed, as well as the different diameters and lengths of the removers according to needs.

The 2 most critical areas where sometimes more power is required are the underlap water course or the overlap, depending on the profile of those areas.

The distributions of the stirrer pins determine an adjustment in the side walls, which ensure that there is no retention of concrete in those walls, remaining always clean.

There is nothing a priori that guarantees the feeding of concrete by the stirrer pins to the pallet, what there is an experience as a base of departure, that allows to propose a good development of work, to find (with the practice in the extrusion head) the best type of stirrer pins shaft for that factory based on its raw materials.